

Observation of anomalously large Nernst effects in conducting polymers

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As a fundamental thermoelectric phenomenon in many solid-state materials, the Nernst effect has yet to be observed in conducting polymers. This knowledge could provide important insight into their elusive mechanism, which are crucial for flexible optoelectronic and thermoelectric applications. However, within the Landau's Fermi-liquid picture, the Nernst coefficient has demonstrated to be proportional to the charge mobility, and thus should be negligible in less ordered polymers with inherent low mobility. Here, we challenge this notion by observing an anomalously large Nernst effect in a range of conducting polymers. Specially, the Nernst coefficients in these doped polymers exceed the Fermi-liquid predictions by 2-3 orders of magnitudes with negative mobility dependence. These intriguing observations are attributed to the intrinsic quasi-one-dimensional transport nature in conjugated polymers and their unique chemical doping mechanism. Our research not only provides experimental insights into the non-Fermi-liquid charge transport nature of polymers, but also suggests its universality for other quasi-one-dimensional materials and/or less ordered systems, and opens up exciting possibilities for developing transverse organic thermoelectric applications.

The Nernst effect is a transverse thermoelectric (TE) effect that describes the generation of a transverse electric field when a longitudinal temperature gradient is applied in the presence of a perpendicular magnetic field (Fig. 1a). Based on Onsager's reciprocity relations, the Nernst effect directly probes the entropy flow of charge carriers, which has provided essential insights into various puzzling solid-state phenomena, such as the pseudo-gap phase and stripe order in high- T_c superconductors^{1–4}, quantum oscillations in topological and low-dimensional materials^{5–7}, phase transition in strange metals^{8,9}, and others^{10–16}. In general, within the framework of Landau's Fermi-liquid (FL) theory, the Nernst coefficient ν is well-defined¹⁷, and is expected to

be proportional to the charge Hall mobility μ_H , and inversely proportional to the Fermi energy E_F ^{8,15–18}, i.e., $\nu_{FL} \propto \mu_H/E_F$ (Fig. 1b). It therefore, predicts negligible Nernst coefficients in low mobility materials, which hampers the basic motivation for investigating the Nernst effect in less ordered systems to both understand their underlying charge transport mechanisms and explore their transverse TE conversion potential.

Conjugated polymers, as an important class of optoelectronic materials, have attracted tremendous attention due to their unique transport properties and potential applications in intrinsically flexible devices^{19,20}. Even with the reports of the prominent Seebeck effect,

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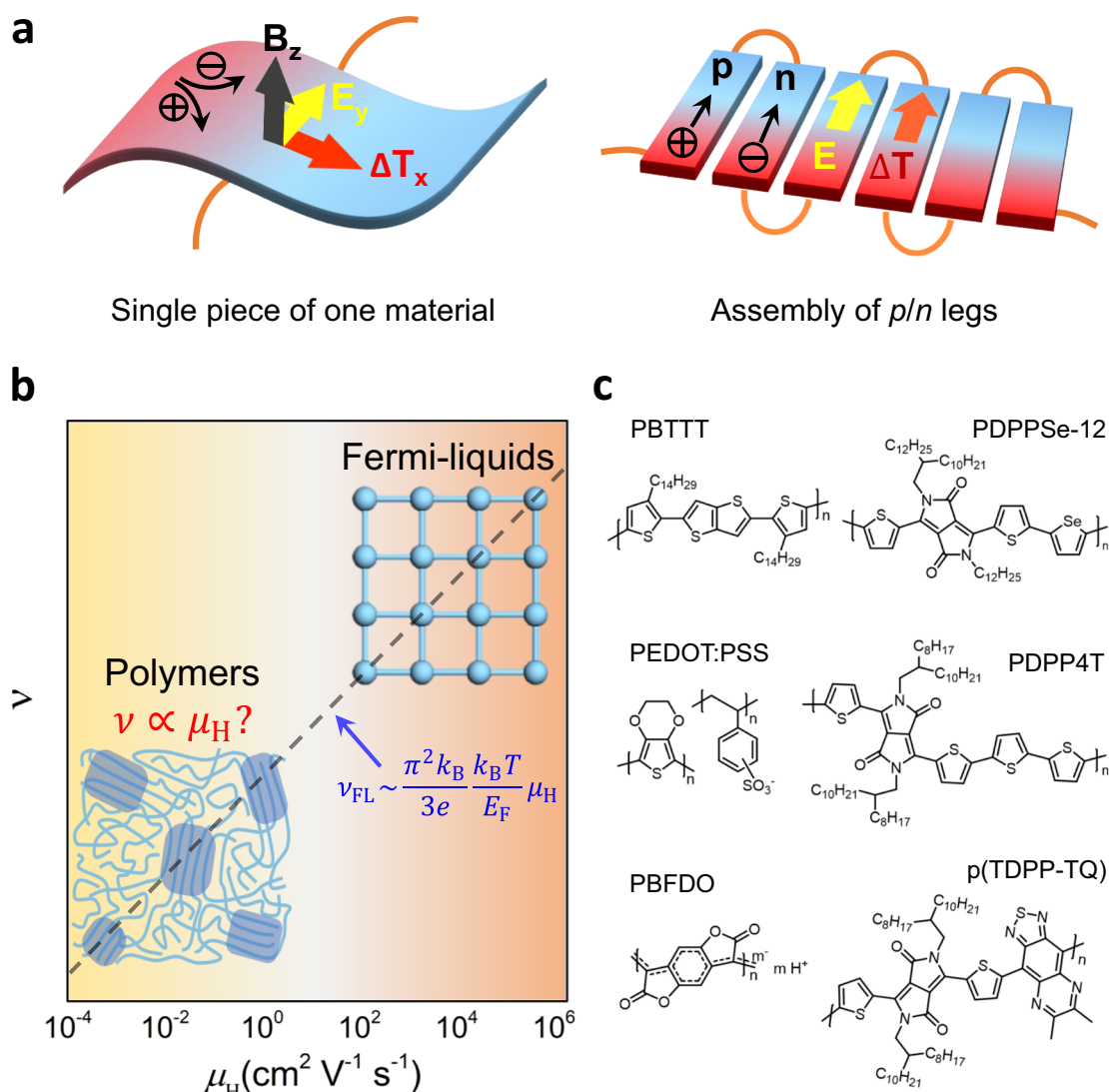


Fig. 1 | Illustration of the Nernst effect and its FL picture. **a** Schematic of a transverse TE device based on the Nernst effect (left), which is a single piece of one material. A transverse electrical field E_y is generated due to the deflection of charge carriers under the presence of a longitudinal temperature gradient ∇T_x and a perpendicular magnetic field B_z . On the right is a schematic of a longitudinal TE device based on the Seebeck effect, which is composed of alternating *p*-type and *n*-type units. **b** Schematic of the expected linear correlation (dashed black line) between the Nernst coefficient ν and charge Hall mobility μ_H within the FL picture,

as indicated by the formula. The bottom left inset qualitatively illustrates that the thin film phase of polymers is typically composed of both crystalline regions and amorphous regions, leading to relatively low mobilities compared to most inorganic crystalline FLs with higher mobilities (top right inset). The left half shading represents a typical mobility range of up to $10 \text{ cm}^2 \text{V}^{-1} \text{s}^{-1}$ for polymers, while the right half shading represents higher mobilities for common crystalline FLs.

c Molecular structures of the as-studied *p*-type and *n*-type polymers.

observations of the Nernst effect in polymeric systems have to the best of our knowledge, not yet been reported. This is mainly because their theoretically predicted ν_{FL} is typically lower than $0.001 \mu\text{V K}^{-1} \text{T}^{-1}$ due to relatively low μ_H ($\sim 0.1\text{--}10 \text{ cm}^2 \text{V}^{-1} \text{s}^{-1}$) and large E_F ($\sim 0.1\text{--}1.0 \text{ eV}$), which is too small to trigger scientific enthusiasm on related topics. In addition, there are significant experimental challenges in detecting the magneto-transport signal in polymers, including the presence of localized charge carriers screening any transverse voltages^{21,22}, a weak signal-to-noise ratio resulting from the inherently low conductivity, and the thermal instabilities in their doped state. However, it is worth noting that conjugated polymers could potentially exhibit more exotic, beyond FL transport mechanism: for example, the inherent, fast one-dimensional (1D) intrachain charge transport that contributes to conduction in between comparatively slow interchain hopping events may be described by Luttinger-liquid (LL) models^{23,24}. This motivates

us to investigate the Nernst effect in conducting polymers to explore their potential non-FL nature and reveal their transverse TE conversion capability.

Herein, we report the general observation of an anomalously large Nernst effect (ALNE) in conducting polymers, which is over 2–3 orders of magnitudes larger than FL predictions. The magnetic field angular dependence measurements verified the ALNE observation. Further doping level dependent characterizations revealed an unusual negative mobility dependence in PBTTT, in contrast to $\nu \propto \mu_H$ in FLs. We propose a phenomenological model of charge scattering and doping in a high doping level regime to explain the observed ALNE response, which could further originate from the inherent quasi-1D transport physics in conducting polymers. Our observations indicate a unique non-FL charge transport picture in conducting polymers, and demonstrate their promising transverse TE conversion potential.

Results and discussions

ALNE Characterization

Hall effect investigation in conducting polymers has been difficult in many cases^{21,22}; the study of the Nernst effect is expected to be even more challenging since it is analogous to a TE Hall effect with a much lower longitudinal current originating from the Seebeck voltage. Furthermore, when a large temperature gradient is applied to enhance the signal, care needs to be taken to avoid dedoping induced instabilities of the thin polymer films. To observe the presumably small Nernst effect in polymers, we aimed to enhance the low signal-to-noise ratio via repeated sweeping of a strong superconducting magnetic field (± 9 T). Furthermore, the Nernst noise is minimized via optimizing the electrical resistance and chemical stability, by manipulating the film thickness, geometry, doping level, patterning, and encapsulation (Supplementary Fig. 1), and by employing a low noise, high-resolution measurement setup ("Method" section). Although various doped polymers have been investigated, we focused on *p*-type doped PBTBT due to its high conductivity and stability in the doped state. To ensure the reliability of the results, the Nernst device and doped PBTBT thin films are systematically characterized. The device temperature gradient is calibrated (Supplementary Fig. 2), and its electrode effect under an external magnetic field is found to be negligible (Supplementary Fig. 3). UV-vis-NIR absorption, UPS, and GIWAXS measurements were performed for the FeCl₃-doped PBTBT films (Supplementary Figs. 4–6), which ubiquitously demonstrated the doping effectiveness and high crystallinity in PBTBT films up to 100 mmol L⁻¹ FeCl₃ doping concentration, consistent with previous PBTBT studies^{22,25}.

In contrast to the tiny Nernst effect predicted by the FL picture, we observed a strong transverse Nernst signal at 300 K in a FeCl₃-doped (80 mmol L⁻¹) PBTBT film (Fig. 2a–c). This transverse Nernst voltage showed a linear dependence on both magnetic field strength (-9 to $+9$ T) and temperature gradient (1 to 5 K), consistent with the typical Nernst effect. The Nernst coefficients $\nu = \frac{V_y L_x}{\Delta T_x B L_z}$ and corresponding Seebeck coefficients $S = V_x / \Delta T_x$ are extracted to be $0.53 \mu\text{V K}^{-1} \text{T}^{-1}$ and $15.7 \mu\text{V K}^{-1}$, respectively, with a $\nu B / S$ ratio of 30.6 % (Fig. 2d). The sign of the observed Nernst effect is negative in the so-called vortex convention¹⁷, indicating larger contributions to the Nernst signal from cold carriers driven by the Seebeck field than hot carriers driven by the temperature gradient. In other words, the energy-dependent mobility $\mu(E)$ must be reduced with increasing carrier energy as more and more holes are induced, i.e., $\frac{d\mu(E)}{dE} < 0$ (Supplementary Fig. 7). In contrast to the transverse Nernst signal, we found the corresponding longitudinal magnetoconductance (MC) and magneto-Seebeck effects were negligible (Supplementary Fig. 8), ruling out the possibility of a transverse signal generated by electrode misalignment. In addition, the possible contribution from the thermal Hall effect was estimated to be negligible (less than 1% of the measured signal). To avoid potential confusion, the notation for temperature is standardized to the italic form *T*, while *T* is used to represent the unit of magnetic field in Tesla.

The most conclusive evidence for the observation of the Nernst effect was its magnetic angular dependency. When the device was rotated in the *xz* plane while keeping the magnetic field fixed along the *z*-axis, ν exhibited a perfect Sine relation with the angular difference θ between the magnetic field and the temperature gradient direction (Fig. 2e). This finding demonstrated the mechanism of the Lorentz force ($\mathbf{v} \times \mathbf{B}$) and confirmed the intrinsic nature of the Nernst effect. Moreover, in addition to the data in Fig. 2 taken at the Institute of Chemistry, Chinese Academy of Sciences (ICCAS), Nernst effect measurements on PBTBT were also carried out independently at the University of Cambridge with a different dopant and the method of ion-exchange doping (1/100 mmol L⁻¹ FeCl₃/BMP-TFSI in acetonitrile)^{25,26}. A consistent magnitude and sign of the Nernst effect were obtained (Supplementary Fig. 9), further ruling out the possibility of potential contributions from the FeCl₄⁻ counterion spins in PBTBT to the

observed Nernst effect, and demonstrating the reliability and reproducibility of our results. This validation is crucial, as the use of pure FeCl₃ for chemical doping could introduce potential interference from the magnetic moments of FeCl₄⁻, as well as chemical degradation of the polymer due to radical side reactions induced by FeCl₃²⁷.

The doping level and temperature dependence of the Nernst effect were further investigated in PBTBT films. Corresponding Seebeck effect, electrical conductivity, and Hall effect measurements were also performed. As the FeCl₃ doping concentration increased from 5 to 100 mmol L⁻¹ at 300 K (Figs. 3a, b and Supplementary Fig. 10), *S* decreased from 30.9 to $15.2 \mu\text{V K}^{-1}$ with the increasing carrier density, while the conductivity σ increased up to a maximum of 589 S cm^{-1} at 40 mmol L⁻¹ doping concentration and then decreased because of over-doping. Meanwhile, μ_H decreased from 1.72 to $0.25 \text{ cm}^2 \text{ V}^{-1} \text{ s}^{-1}$. These trends are consistent with typical chemical doping effects in polymers. Notably, for single band polymers, the extracted μ_H here are most likely underestimated compared to the intrinsic drift mobility due to incoherent transport and the screening effect that localized hopping carries may exert on the transverse voltage²⁸. However, we argue that μ_H should still be considered to be the relevant, effective mobility for the Nernst effect and predicting ν_{FL} , since the Nernst and Hall effects are both transverse transport phenomena originating from the Lorentz force acting on the same charge carriers. Any physical process, such as screening, that might affect/reduce the Hall coefficients, should also affect the Nernst coefficients.

Surprisingly, there was an enhancement in ν from 0.41 to $0.69 \mu\text{V K}^{-1} \text{T}^{-1}$ with increasing doping concentration. This resulted in a negative dependence of the Nernst coefficient on the measured Hall mobility and Seebeck coefficient, which is contrary to the expected behavior within the FL picture. To ensure the reliability of the results, in-situ studies were conducted by performing dedoping in a 100 mmol L⁻¹ FeCl₃-doped PBTBT film (Supplementary Fig. 11). The results further confirmed the unusual doping level dependence of the Nernst effect in polymers. Figure 3c plots the temperature dependence of ν and *S* in a 100 mmol L⁻¹ FeCl₃-doped PBTBT (Supplementary Fig. 12). The applied temperature gradient at different temperatures is plotted in Supplementary Fig. 2d. The linear increase of *S* with temperature aligns with the free-electron model and indicates a typical metallic behavior, while the non-linear temperature dependence of ν is more complicated and has been reported in different materials^{8,14}. The corresponding normalized temperature dependent σ and μ_H of PBTBT are plotted in Fig. 3d, both showing a decrease at cryogenic temperatures. However, at temperatures above 200 K, they are nearly unchanged, indicating the onset of metallic transport and in accordance with the reported coherent transport in doped PBTBT films^{22,25}. In addition, temperature-dependent MC and magnetic susceptibility χ measurements were performed for doped PBTBT films (Supplementary Figs. 12 and 13). These measurements revealed a Pauli spin susceptibility with a linear relation of $\chi T \sim T$ and an enhanced charge phase-coherence length at cryogenic temperatures, further supporting a coherent charge transport picture.

The Nernst effect in PBTBT is quantitatively compared with the predicted $\nu_{\text{FL}} = 283 \frac{\mu_H}{T_F} T$ (Fig. 3e)^{8,15–18}, where *T* is the measurement temperature and $E_F = k_B T_F$. In Fig. 3e, the PBTBT Hall effect mobility is applied (Fig. 3b), and the E_F values can be determined from the in-situ measured Seebeck coefficients (Fig. 3a), under rigid band and weak energy derivative of transport function approximation²⁹. Specifically, in the framework of a Fermi gas, free-electron model, the Seebeck coefficients are inversely proportional to E_F , expressed as $S = \frac{\pi^2 k_B^2}{3e} \frac{T}{E_F}$. Although the free-electron model may not fully capture the intricate behavior of doped polymers, it serves as a suitable approximation. This is supported by its alignment with the linear temperature dependence of the Seebeck coefficient (Fig. 3c), the measured Pauli spin susceptibility (Supplementary Fig. 13), and the presence of highly delocalized charge carriers in heavily doped PBTBT, which suggests a coherent,

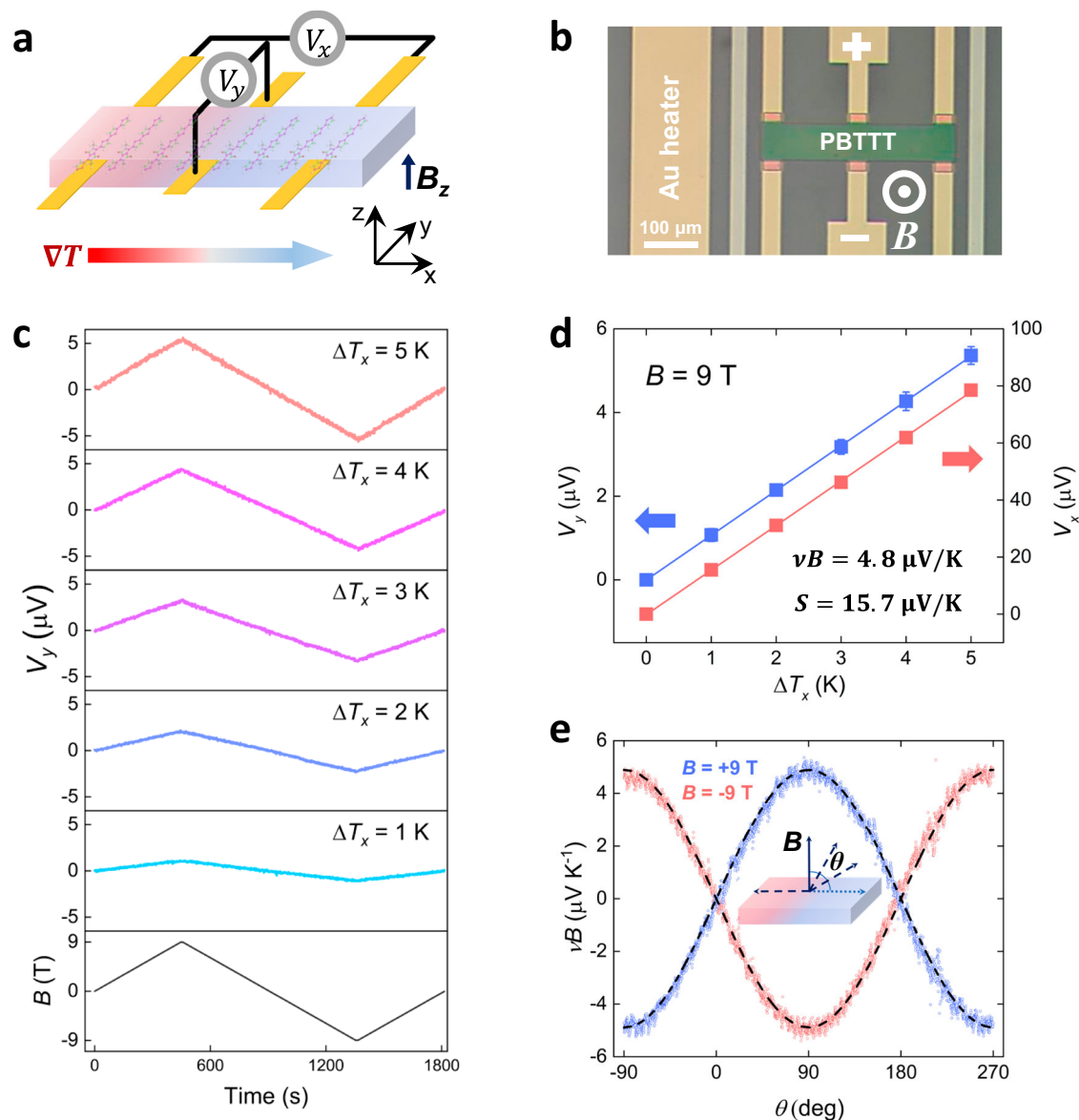


Fig. 2 | Nernst effect in doped PBTTT at 300 K. **a** Schematic diagram and **(b)** photograph of the PBTTT Nernst device. The Seebeck (V_x) and Nernst (V_y) effects were measured in-situ simultaneously with an in-plane temperature gradient ∇T established by the Au heater. An out-of-plane magnetic field B_z is applied for the Nernst effect measurements, in which the presented direction is defined as positive. The plus and minus signs on the electrodes denote the wiring setup of the Nernst signal measurements. The film aspect ratio is set by $L_x/L_y = 4.5$. **c** Measured transverse Nernst voltage signal V_y versus magnetic field under different temperature gradients in an 80 mmol L⁻¹ FeCl₃-doped PBTTT film. **d** Extracted Seebeck

(red line) and Nernst (blue line) voltage signals when $B = 9$ T versus different applied temperature gradients. **e** Measured Nernst coefficient versus the angle θ between the rotating temperature gradient ($\Delta T = 4$ K) and the perpendicular magnetic field ($B = -9$ T and 9 T), as the device is rotated in the xz plane while the magnetic field is fixed in the z -direction. The dashed black curve plots the Sine function according to the Lorentz force. Error bars were determined from uncertainty in the extraction of electromotive force and represent one standard deviation. Source data are provided as a Source Data file.

metallic charge transport mechanism as previously demonstrated. Consequently, this approach yields E_F values in the range of 0.24–0.48 eV for doped PBTTT.

To validate these E_F values, two alternative methods were applied. First, magnetic susceptibility characterizations are applied by plotting χT versus T as $\chi T = \chi_{\text{Pauli}} T + C$, to extract the free-electron Pauli contribution χ_{Pauli} (Supplementary Fig. 13). The potential contribution of FeCl₄⁻ counterions to the Curie susceptibility was investigated and ruled out (Supplementary Fig. 14). The density of states (DOS) at Fermi energy was then calculated by $\rho(E_F) = \chi_{\text{Pauli}} / \mu_B^2$. Since the as-studied doped PBTTT thin films can be physically recognized as disordered metals, from the perspective of FL theory and electron filling model for typical 3D metallic systems, their E_F values can be estimated as

$\rho(E_F) = \frac{m^* \sqrt{2m^*}}{\pi^2 \hbar^3} \sqrt{E_F}$. This leads to E_F values ranging from 0.082 to 0.21 eV for PBTTT with FeCl₃ doping concentration of 5 to 100 mmol L⁻¹ (Supplementary Table 1), respectively. To be conservative with the E_F values, we applied a larger effective mass m^* of 1.61 times free electron mass m^0 along the π - π stacking direction, compared to the effective mass of 0.1 m^0 along the polymer chain direction³⁰. In addition, by definition, the E_F can be directly extracted from previously well-studied intrinsic and doped PBTTT band diagrams and density of states calculations. Across various references and different theoretical models of PBTTT structure^{30–33}, the extracted E_F values are consistently around 1 eV (Supplementary Table 2).

The E_F values determined from the three different methods are consistent with each other, all are on the order of 0.1–1 eV.

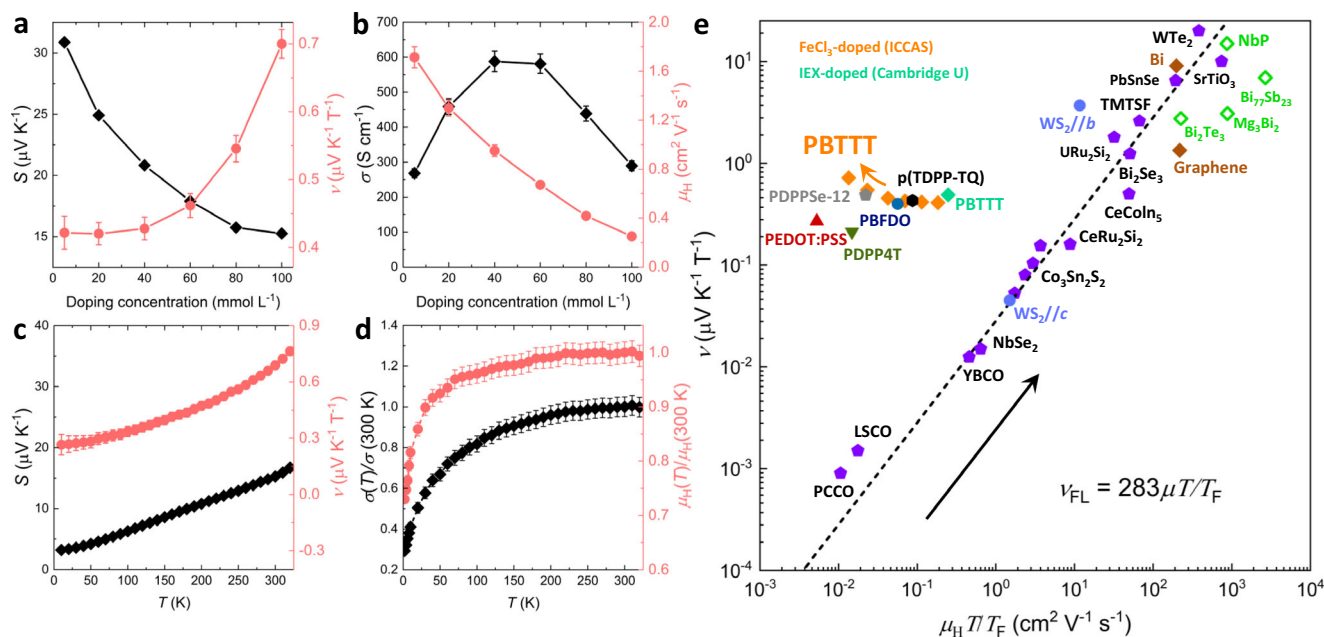


Fig. 3 | Doping level and temperature dependence of ALNE in PBTTT films, as well as ALNE in different polymers and comparison to other materials. **a** PBTTT thermopower S and Nernst coefficients ν versus different FeCl_3 doping concentration. **b** Doping concentration dependent PBTTT electrical conductivity σ and Hall mobility μ_H . **c** In-situ measured temperature T dependence of S and ν in one 100 mmol L^{-1} FeCl_3 -doped PBTTT device. **d** Corresponding temperature dependence of σ and μ_H of (c), which are normalized to their values at 300 K. The error bars in μ_H , S and ν were determined from the uncertainty in the extraction of the electromotive force and represent one standard deviation. The error bars in σ were determined from the uncertainty in the film thickness measurements and represent one standard deviation. **e** Plot of Nernst coefficients ν versus $\mu_H T/T_F$ for different polymers, bismuth, and graphene measured in this work at 300 K and various other materials reported in literature. T_F is the material Fermi temperature. The dashed black line represents the theoretical relationship predicted in the

framework of FL theory. The presented polymer data points are located at the top-left area, denoted by various solid symbols with different colors, and largely deviate from the FL line, while the bismuth and graphene data points are denoted as brown solid diamonds and qualitatively follow the FL line. The ν , μ_H , and T_F of IEX-doped PBTTT were measured independently at the University of Cambridge (blue-green solid diamond), while the measurements of FeCl_3 -doped PBTTT and other materials were performed at ICCAS. The direction of the orange arrow indicates the increasing FeCl_3 doping concentration of PBTTT. The green data points represent materials with the Nernst effect reported near or at room temperature. The Nernst effect of materials represented by purple points and WS_2 are reported at low temperatures. The detail information of ν , μ_H , T , and T_F for the data points shown are summarized in Supplementary Table 3. Source data are provided as a Source Data file.

By combining the Seebeck determined E_F with the μ_H of $0.25\text{--}1.72\text{ cm}^2\text{V}^{-1}\text{ s}^{-1}$, the predicted ν_{FL} of PBTTT doped with $5\text{--}100\text{ mmol L}^{-1}$ FeCl_3 are determined to be $0.0052\text{--}0.00038\text{ }\mu\text{V K}^{-1}\text{ T}^{-1}$ at 300 K, as compared to our experimental ν values of $0.41\text{--}0.69\text{ }\mu\text{V K}^{-1}\text{ T}^{-1}$ (Fig. 3a), respectively. For the 100 mmol L^{-1} FeCl_3 doped PBTTT, the measured ν is over 3 orders of magnitudes larger than the predicted ν_{FL} . A stress test was applied by replacing the as-extracted μ_H with the corrected maximum PBTTT charge mobility μ_d of $8.0\text{ cm}^2\text{V}^{-1}\text{ s}^{-1}$ determined by a refined analysis of the Hall effect that considers screening effects (Supplementary Fig. 15)²¹. Even under this conservative scenario, the extracted ν is still over 1 order of magnitude larger than the predicted ν_{FL} .

In addition to the independent measurements of FeCl_3 -doped PBTTT at ICCAS and ion-exchange (IEX) doped PBTTT at Cambridge, a negative Nernst effect with similar magnitude was also observed in various other polymer thin films (Fig. 3e and Supplementary Figs. 16 and 17), including PBFDO³⁴, PEDOT:PSS, PDPP4T, PDPPSe-12³⁵, and p(TDPP-TQ)³⁶, which have different molecular and thin film microstructures. Some of the polymers were also n -type doped. The extracted ν are compared with the predicted ν_{FL} in the same approach as PBTTT. In Fig. 3e, we also present some of the reported room temperature (green dots) and low temperature (purple dots) ν values in a variety of materials (Supplementary Table S3)^{8,15–18,37–41}. Remarkably, the measured ν values in polymers are generally 2 to 3 orders of magnitudes larger than ν_{FL} . In sharp contrast, most other materials, including inorganic narrow-gap semiconductors and semimetals, superconductors, topological matters, two-dimensional systems, and

even organic crystals of TMTSF derivatives, show a generally good agreement between ν and the estimated $\mu_H T/T_F$ scaling relationship over 6 orders of magnitudes and are consistent with the ν_{FL} typically within an order of magnitude. Notably, many reported ν values at room temperature are even smaller than the corresponding FL predictions. To further confirm the anomaly of the observed polymeric Nernst effect, we also characterized as part of our study the widely investigated graphene and polycrystalline bismuth thin films at 300 K (brown dots in Fig. 3e and Supplementary Figs. 18 and 19). The results indicate that their ν values are consistent with previously reported measurements^{7,42–44}, as well as with the FL predictions, in contrast to the polymers.

Mechanism

Within the FL picture, in addition to large μ_H/E_F , some of the reported large Nernst effect can be attributed to multi-band transport or phonon contributions^{14,45,46}. The observed polymeric Nernst effect reveals a monotonic decrease with temperature, thus ruling out the possibility of phonon drag which would typically increase at cryogenic temperatures. Multi-band transport has been invoked to explain some of the transport signatures of some donor-acceptor conducting polymers⁴⁷. However, for PBTTT, there have been no reported experimental evidence for such ambipolarity and multi-band transport behavior so far. In the temperature range of interest here we also do not observe significant deviations from a linear temperature dependence or any sign changes in the Seebeck coefficient (Fig. 3c), that could be manifestations of ambipolar transport. Even for the strange

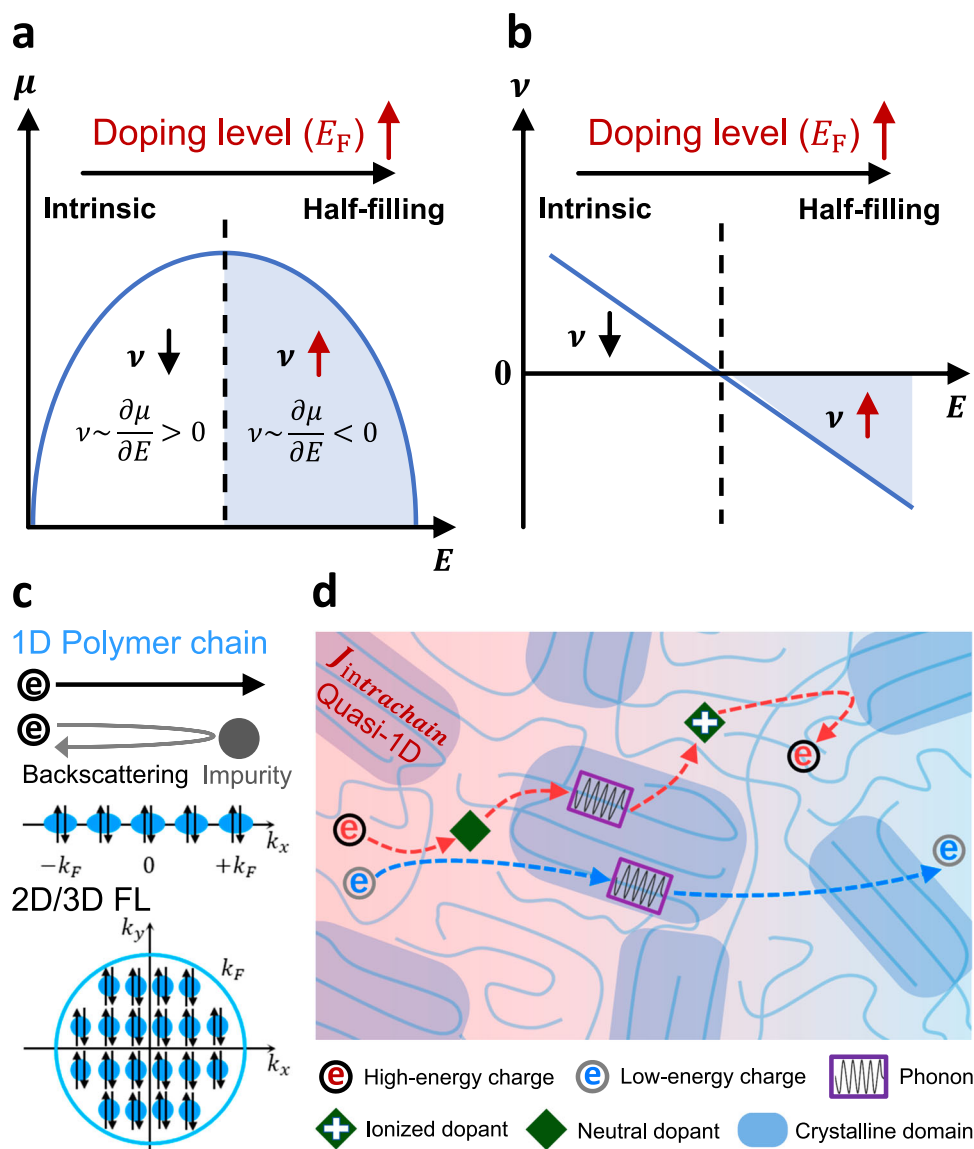


Fig. 4 | Mechanism for the observed negative ALNE in doped polymers.

a Schematic of typical doping level and Fermi energy dependence of charge mobility in conducting polymers from intrinsic (undoped) to the highest doping level (half-filling) state. The shaded area represents the regime for negative Nernst coefficients ν . The down and up arrows indicate the decrease of positive and the increase of negative ν magnitude. The ν sign transition point is indicated by the

dashed black line. **b** Schematic of the doping level dependence of the Nernst coefficients in conducting polymers corresponding to **(a)**. **c** Illustration of the difference in Fermi surfaces and scattering physics for 1D polymer chain (up) and 2D/3D FLs (down). **d** Schematic of diverse and strongly energetic dependent quasi-particle scattering nature in inherently disordered polymers with high doping level.

metal state of 2M-WS₂//b shown in Fig. 3e, which is beyond the boundary of the FL picture and exhibits a maximum ν at 25 K around its quantum critical point⁸, its Nernst coefficient is only nearly one order of magnitude larger than the predicted ν_{FL} due to peculiar strange metal physics. This discrepancy with the ν_{FL} is less dramatic compared to the observed ALNE in polymers. Furthermore, strange metals still qualitatively follow the positive correlation between ν and μ_H , in contrast to the observed unusual doping level and mobility dependence of ν in PBTtT (Fig. 3e). So far, there have been no documented studies that can understand the observed ALNE, which indicates an unusual mechanism in the case of polymers.

We first discuss the doping level dependence of the Nernst coefficient expected for PBTtT. As indicated in Fig. 4a, at low doping level, the charge mobility of polymers should increase with increasing doping level due to reduced Coulomb trapping and decreased π - π stacking distance (Supplementary Fig. 6), which will facilitate more

efficient interchain hopping transport. In this low doping level regime, the Nernst coefficient should be positive, as $\nu \sim \frac{\partial \mu(E)}{\partial E} > 0$. As the doping level continuously increases and polymers approach the half-filling state, where the density of states is at a maximum, the mobility will reach a maximum and then start to decrease. When $\frac{\partial \mu(E)}{\partial E}$ turns negative, the Nernst coefficient should also become negative. Our experiments, however, failed to observe a positive-to-negative sign transition as a function of doping level, most likely due to experimental limitations. At very low doping levels with low conductivities the Nernst coefficient is too difficult to measure due to the high resistance of the channel. In the regime accessible by the FeCl₃ or IEX-doping, all our PBTtT films were already relatively highly doped, even with the lowest FeCl₃ doping concentration of 5 mmol L⁻¹. In this regime, the charge mobility reduces with increased doping level as shown in Fig. 3b, thus producing a negative Nernst coefficient due to $\frac{\partial \mu(E)}{\partial E} < 0$, which magnitude increases with the doping

level and thus explains our observation of the unusual doping level dependence of the ν in PBTTT (Fig. 4b).

Regarding the anomalously large magnitude of the observed Nernst effect in doped polymers as well as the breakdown of FL theory, this could be related to their inherent quasi-1D transport nature. 1D systems are described by the Tomonaga-Luttinger liquid (LL) model. LLs are expected to exhibit unusually strong scattering mechanisms and quasiparticle interactions due to the confinement in the other two dimensions (Fig. 4c)^{48,49}, thus could theoretically produce anomalously large $\frac{\partial \mu(E)}{\partial E}$ and Nernst effects, as the Nernst coefficient measures the carrier transport entropy which can be directly gained from the charge scatterings. Previous studies have already indicated various anomalous thermal transport behaviors in 1D systems^{23,50–54}, including significantly enhanced Nernst effect predicted in weakly-interacted 1D chains⁵³. An intriguing Nernst response has also been observed in the metallic states of quasi-1D organic charge transfer salts when the magnetic field is oriented along their magic angle⁵⁵. In addition, in chemically doped polymers, other scattering mechanisms and quasiparticle interactions could be relevant, such as poorly screened Coulomb, carrier-carrier interactions, strong electron-phonon couplings, or energy filtering by the grain boundaries, reflecting the relatively high charge density and disordered nature of these materials (Fig. 4d), in contrast to most weakly interacting FLs. In particular, these charge scatterings could be strongly energy dependent in doped polymers, i.e., larger $\frac{\partial \mu(E)}{\partial E}$, which will essentially enhance the carrier transport entropy and the Nernst effect.

Such a postulated, strong energy-dependence of charge scattering predicted by LL physics, as well as the intrinsic disorder nature of doped polymers could explain the failure of FL predictions regarding the size of the Nernst effect. It is also consistent with the unusual doping level and charge mobility dependence observed in PBTTT, as the PBTTT doping level and charge density increase, its structural disorder is clearly enhanced as evidenced by the GIWAXS pattern revealing reduced crystallinity along both lamella and π - π directions (Supplementary Fig. 6) as well as smaller Hall mobilities, which will lead to more pronounced charge scattering. Meanwhile, the enhanced structural disorder will weaken the interchain coherence, enabling the crystalline regime to approach a true 1D state. These combined lead to a larger Nernst effect at a higher PBTTT doping level. In fact, Fig. 3e already suggests that among different polymers, the ALNE will be more dramatic with the increase of disorder and charge density, as the amorphous PEDOT:PSS exhibits the largest difference between the ν and predicted ν_{FL} . To further explore the origin of the observed ALNE, we examined the molecular weight dependence of the PBTTT Nernst effect (Supplementary Fig. 20). The results show a significantly larger Nernst effect with increasing molecular weight, likely due to enhanced quasi-1D intrachain charge transport, which supports the proposed mechanism.

In principle, conducting polymers can be considered as highly anisotropic, prototypical 1D systems. Polymer films are typically composed of ordered, quasi-crystalline regions with aligned 1D chains, which are separated by more disordered, amorphous regions with disordered chains. Within the crystalline regions, both strong intrachain and weaker, but finite interchain interactions coexist, while the amorphous regions most likely mainly support isotropic intrachain transport due to much less pronounced π - π stacking. Intrachain transport in the crystalline regions can be considered as quasi-1D^{23,24}, but previous studies have shown that the finite interchain interactions, driven by π - π stacking, induce a 2D (or 3D depending on the nature of interchain packing) electronic structure. This results in a crossover from a true 1D LL to a 2D/3D FL state for the overall charge transport in coupled polymer chains. It, therefore, remains difficult to accurately assess the relevance of LL physics to explain our ALNE observations in polymers, since a comprehensive theoretical framework for describing the charge carrier magneto-transport in such quasi-1D systems in the presence of significant disorder is not yet available.

In this study, we present the general observation of ALNE in conducting polymers in which negative Nernst coefficients exceed the predictions of FL theory by several orders of magnitudes with unusual doping level dependence in PBTTT. Our observation indicates greatly enhanced charge transport entropy with negative $\frac{\partial \mu(E)}{\partial E}$ in highly doped conducting polymers, which we attributed to strong energy dependence of charge scatterings and could be further originated from the inherent quasi-1D LL physic in conducting polymers. The general ALNE response provides experimental insights into the non-FL nature of charge transport and TE conversion mechanism in conducting polymers and other quasi-1D materials. We anticipate that our study will invigorate future theoretical studies for a comprehensive physical framework of magneto-thermoelectric transport phenomena in less ordered and/or one-dimensional systems. Furthermore, it might open up possibilities for the development of advanced transverse organic TE materials for state-of-the-art wearable electronics.

Methods

Materials

PBTTT-C14 (Sigma-Aldrich); PBFDO (Volt-Amp Optoelectronics); PEDOT:PSS (Clevios™, Heraeus); PDPP4T (Organtec Ltd.); PDPPSe-12 was synthesized following ref. 35; p(TDPP-TQ) was synthesized following ref. 36; iron (III) chloride (97%, Sigma-Aldrich); o-dichlorobenzene (anhydrous, 99%); chloroform (HPLC, Concord); nitromethane (anhydrous, >98.5%, Sigma-Aldrich); 1-butyl-1-methylpyrrolidinium bis(trifluoromethanesulfonyl) imide (BMP-TFSI, 99.9%, Solvionic); acetonitrile (>99.9%, Romil Ltd).

Device fabrication

The commercially bought PEDOT:PSS and PBFDO solutions were directly used without further treatments. Other polymer solutions were prepared at ambient temperature in a nitrogen glovebox. PBTTT, PDPPSe-12, and p(TDPP-TQ) were heated to 100 °C to dissolve in o-dichlorobenzene at a concentration of 10 mg/ml, while the PDPP4T was dissolved in chloroform also with a concentration of 10 mg/ml. The Au and Pt electrodes of Nernst substrates were patterned by lithography and sputtering on clean glass substrates in advance. After patterning, the substrates were cleaned first by ultrasonication in isopropanol, acetone, and deionized water, and then by O₂ plasma for 15 minutes. The PEDOT:PSS, PBTTT, and PBFDO thin films were fabricated by spin-coating onto the patterned Nernst substrates with a typical thickness of 50 nm, 50 nm, and 200 nm, respectively. The PDPP4T, PDPPSe-12, and p(TDPP-TQ) films were fabricated by drop-casting onto clean Nernst substrates with a typical thickness of around 1.5–6 μ m. The polymer films were then annealed in a nitrogen atmosphere with the following conditions: PBTTT 2 hours at 180 °C; PDPPSe-12 1 h at 180 °C; p(TDPP-TQ) 1 hour at 150 °C; PBFDO, PEDOT: PSS, and PDPP4T 1 h at 100 °C. The as-annealed polymer films were then patterned by standard lithography procedures for accurate Nernst measurements. After patterning, subsequent sequential doping was achieved chemically by immersing the films in FeCl₃/CH₃NO₂ solution for PBTTT, PDPP4T, PDPPSe-12, and p(TDPP-TQ) at ambient temperature in a nitrogen atmosphere with different concentrations for one and half minutes. For ion-exchange doping, the polymer films were immersed in the dopant solution (1/100 mM FeCl₃/BMP-TFSI in acetonitrile) at ambient temperature in a nitrogen atmosphere for one minute. The as-fabricated PEDOT:PSS and PBFDO devices were already in the doped state. The films were then encapsulated by a spin-coated Cytop layer with a typical thickness of 300 nm. Finally, the Nernst devices were fixed onto standard PPMS pucks and connected by spot welding using Al wires.

PPMS measurements

Measurements of electrical conductivity, Seebeck, Hall, Nernst effect, and magnetic susceptibility were carried out in-situ under helium gas (5–10 Torr) with a Quantum Design Physical Property Measurement

System (PPMS, DynaCool-9T) which is located in a quiet and clean room. The input electrical signals to the device are applied by an Agilent B2902A source unit and the resulting signals are measured by Keithley 2182A nanovoltmeters with an input impedance of 10 GΩ. All the wirings outside the PPMS chamber were custom made low-noise coaxial cables, which results in an overall electrical noise of less than 10 nV when the system is shorted. For TE measurements, the temperature gradient is created by applying a voltage to the Au heating electrode on the Nernst device in constant voltage mode, and the calibration of temperature difference is performed by the two Pt thermoresistors within the device channel, in which one on the hot side while the other one is on the cold side. Detailed temperature calibration procedures and data are presented in supplementary information. The magnetic field applied in DC Hall and Nernst measurement is the PPMS built-in ± 9 T superconducting magnetic field, which is normally in the perpendicular direction while the device can be rotated 360°. For temperature-dependent experiments, the environment temperature of PPMS can be changed from 2–400 K. To achieve a clean Nernst effect signal in polymer thin films at 300 K, the Nernst device channel resistance typically needs to be below 5 kΩ, alongside an electrical noise level of less than 100 nV. Moreover, for polymer Nernst effect measurements at 300 K, the applied temperature gradient should be smaller than 10 K to avoid significant dedoping. A typical Nernst effect measurement would last about one hour for the applied field sweeping speed of 200 Oe/Sec.

Electrical conductivity

The polymer device electrical conductivity is measured via a standard four probe Van der Pauw method. The film thickness is measured by a Dektak XT step profiler.

Hall effect

The Hall coefficient R_H and Hall mobility μ_H are measured by applying a constant longitude current I to the polymer film while the transverse Hall voltage signal V_H is measured under perpendicular magnetic field. R_H and μ_H are calculated in a standard manner:

$$R_H = \frac{V_H d}{IB} \quad (1)$$

$$\mu_H = |R_H| \sigma \quad (2)$$

Seebeck effect

The device thermopower S are measured in a longitude setup with an established temperature difference ΔT , by definition:

$$S = \frac{V_x}{\Delta T_x} \quad (3)$$

Nernst effect

For Nernst effect measurements, the transverse Nernst voltage V_y is measured with an established longitude temperature gradient under a perpendicular magnetic field. The Nernst coefficient ν is calculated via:

$$\nu = \frac{E_y}{\nabla T_x B} = \frac{V_y}{\Delta T_x B} \frac{L_x}{L_y} \quad (4)$$

where, $L_x = 450 \mu\text{m}$ and $L_y = 100 \mu\text{m}$ for the presented device geometry.

SQUID measurements

The magnetic susceptibility measurements were carried out by the PPMS Vibrating Sample Magnetometer (VSM) system with a

temperature range of 2–300 K under a constant magnetic field of 1 T. Nonmagnetic silicon substrates were employed. The data were further corrected with bare substrates.

UV-vis-NIR absorbance, UPS, and GIWAXS measurements

The in-situ UV-vis-NIR measurements were carried out with a Shimadzu UV3600plus system. The UPS measurements were conducted by a Kratos ULTRA AXIS DLD ultra-high vacuum photoelectron spectroscopy system with a He discharge lamp (He I) as the excitation source. The GIWAXS measurements were conducted by an Anton Paar SAX-Spoint 5.0 system with an X-ray wavelength of 1.5418 Å and 0.14° as an incidence angle. The diffraction intensity was detected with a 2D HPC Detector (PILATUS3 R 1 M). All X-ray measurements were performed in a vacuum environment to minimize air scattering and beam damage to samples.

Data availability

The data that support the findings of this work are available within the paper and the supplementary information. Source data are provided in this paper.

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Author contributions

C.D. and Y.M. conceived the idea. C.D., H.S., and D.Z. supervised the project. Y.M. conceived the experiments, performed the transport measurements at ICCAS, and analyzed the results. Y.Z. and W.Z. observed the initial Nernst phenomenon. X.R. and Y.M. performed the transport measurements in Cambridge. Y.M., Z.J., C.Y., X.D., F.Z., and Y.Z. performed the GIWAXS, UV-vis-NIR, VSM, and UPS measurements. J.F. helped to fabricate PBTBT thin films. G.G. and T.L. synthesized the p(TDPP-TQ) polymer. D.W. and L.X. helped to analysis the data. Y.M., Y.Z., C.D., and H.S. wrote the manuscript. All authors discussed the results and contributed to the preparation of the final draft.

Competing interests

The authors declare no competing interests.

Additional information

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